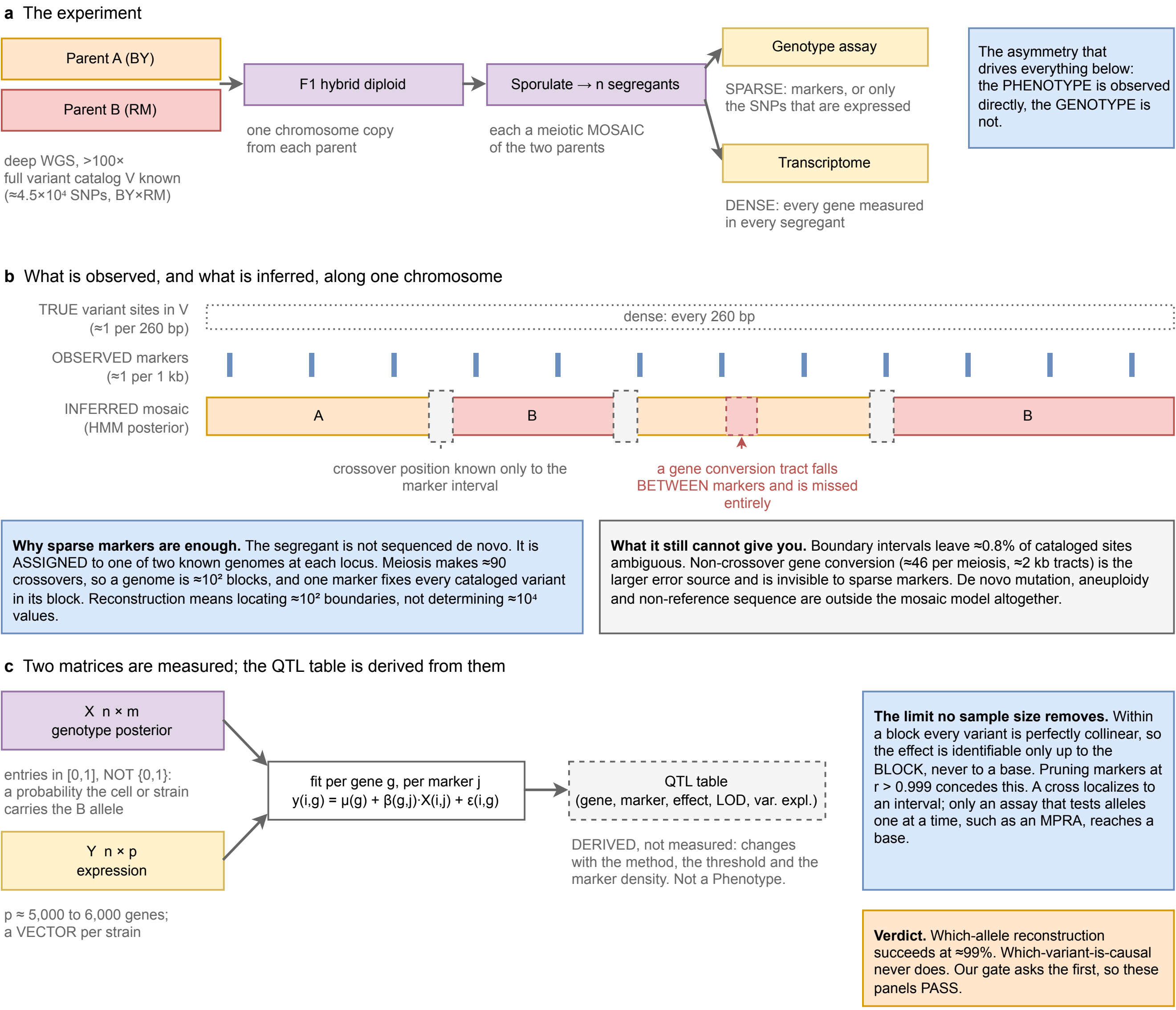
a The experiment
Parent A (BY)
Parent B (RM)
deep WGS, >100×
full variant catalog V known
(≈4.5×10^4 SNPs, BY×RM)
F1 hybrid diploid
one chromosome copy
from each parent
Sporulate → n segregants
each a meiotic MOSAIC
of the two parents
Genotype assay
SPARSE: markers, or only
the SNPs that are expressed
Transcriptome
DENSE: every gene measured
in every segregant
The asymmetry that drives everything below: the PHENOTYPE is observed directly, the GENOTYPE is not.
b What is observed, and what is inferred, along one chromosome
TRUE variant sites in V
(≈1 per 260 bp)
dense: every 260 bp
OBSERVED markers
(≈1 per 1 kb)
INFERRED mosaic
(HMM posterior)
A
B
B
crossover position known only to the
marker interval
a gene conversion tract falls
BETWEEN markers and is missed
entirely
Why sparse markers are enough. The segregant is not sequenced de novo. It is ASSIGNED to one of two known genomes at each locus. Meiosis makes ≈90 crossovers, so a genome is ≈10^2 blocks, and one marker fixes every cataloged variant in its block. Reconstruction means locating ≈10^2 boundaries, not determining ≈10^4 values.
What it still cannot give you. Boundary intervals leave ≈0.8% of cataloged sites ambiguous. Non-crossover gene conversion (≈46 per meiosis, ≈2 kb tracts) is the larger error source and is invisible to sparse markers. De novo mutation, aneuploidy and non-reference sequence are outside the mosaic model altogether.
c Two matrices are measured; the QTL table is derived from them
X n × m
genotype posterior
entries in [0,1], NOT {0,1}:
a probability the cell or strain
carries the B allele
Y n × p
expression
p ≈ 5,000 to 6,000 genes;
a VECTOR per strain
fit per gene g, per marker j
y(i,g) = μ(g) + β(g,j)·X(i,j) + ε(i,g)
QTL table
(gene, marker, effect, LOD, var. expl.)
DERIVED, not measured: changes
with the method, the threshold and the
marker density. Not a Phenotype.
The limit no sample size removes. Within a block every variant is perfectly collinear, so the effect is identifiable only up to the BLOCK, never to a base. Pruning markers at r > 0.999 concedes this. A cross localizes to an interval; only an assay that tests alleles one at a time, such as an MPRA, reaches a base.
Verdict. Which-allele reconstruction succeeds at ≈99%. Which-variant-is-causal never does. Our gate asks the first, so these panels PASS.